

Multi-method consensus analysis of targeted *in situ* spatial transcriptomics: a framework with pilot demonstration on human lung adenocarcinoma

Miguel Ángel Díaz-Campos¹ and Alfredo de Jesús Rodríguez Gómez^{1,†}

¹Instituto Nacional de Pediatría (INP), México City, México

[†]Corresponding author: alfredo.rodriguez@iibiomedicas.unam.mx

Abstract

Spatial transcriptomics technologies enable simultaneous profiling of gene expression and cellular organization within intact tissues, yet no consensus methodology exists for integrating multiple analytical frameworks into reproducible biological inference. Here we present a multi-tool consensus pipeline for *in situ* spatial transcriptomics and apply it to a publicly available Xenium dataset of human lung invasive acinar adenocarcinoma (268,034 cells, 289 genes; one tissue section, one patient). For each analytical family — annotation, spatially variable gene detection, spatial domain inference, cell–cell communication, and pseudotime trajectory — we apply at least two independent computational tools and retain only findings concordant across methods. The framework identifies 288 of 289 genes with significant spatial autocorrelation across three orthogonal tests, two spatial domain partitionings (Banksy and Novae) with 35.4% high-confidence cell-level agreement (the remaining 64.6% representing boundary or transition states between domains), and 20 of 30 candidate cell–cell communication interactions retained under both

Benjamini–Hochberg-corrected ligand–receptor scoring and Bonferroni-corrected Bayesian spatial inference. The dominant validated hub features the sheddase ADAM17 and the oncomucin MUC1, with co-localization signal across six independent myeloid and stromal sender populations converging on tumor cells (Spacia p -values from 4.97×10^{-31} to 1.10×10^{-71}). Within macrophages, a focused sub-trajectory inferred independently by diffusion pseudotime and Slingshot (Methods) reveals a continuum of activation states — not a directional M1→M2 polarization, as the canonical M1 and M2 markers both increase along the inferred pseudotime — accompanying this signaling architecture. These results, derived from a single tissue section, establish a reproducible analytical framework for targeted spatial transcriptomics; biological generalization across lung adenocarcinoma subtypes and stages will require validation in patient cohorts.

Contents

Introduction

The development of *in situ* spatial transcriptomics platforms has transformed tissue biology by enabling simultaneous measurement of gene expression and spatial context at single-cell resolution [Moses and Pachter, 2022]. Among these technologies, the 10x Genomics Xenium platform provides subcellular transcript detection across targeted gene panels of 100–5,000 genes, making it suitable for rapid, cost-effective clinical tissue profiling [Janesick et al., 2023]. However, the targeted nature of such panels — typically 289–480 genes — poses significant analytical challenges: dimensionality-reduction and clustering methods developed for whole-transcriptome data may underperform, cell-type annotation becomes marker-limited, and spatially variable gene detection must operate on a pre-selected gene universe.

A fundamental, underappreciated challenge in spatial transcriptomics analysis is methodological discordance. Independent computational tools applied to the same dataset routinely produce different cell-type assignments, different sets of spatially variable genes, and different cell–cell communication predictions [Dimitrov et al., 2022, Efremova et al., 2020]. Without a principled strategy for reconciling these discrepancies, it is difficult to distinguish biologically robust signals from tool-specific artifacts. Consensus approaches — requiring independent confirmation from at least two orthogonal methods — have been advocated in bulk and single-cell omics but have not been systematically applied to spatial transcriptomics workflows [Luecken and Theis, 2019].

Lung adenocarcinoma represents a compelling biological context for such a framework. The tumor microenvironment (TME) of lung adenocarcinoma is characterized by complex cellular crosstalk between malignant epithelial cells, myeloid populations, and adaptive immune subsets, yet the spatial organization of these interactions remains incompletely understood [Laughney et al., 2020, Wu et al., 2021]. In particular, macrophage polarization states and their ligand–receptor interactions with tumor cells have emerged as key determinants of immunotherapy response [Cassetta and Pollard, 2020], but direct spatial evidence for dominant signaling axes in intact FFPE tissue has been limited by the scarcity of spatially resolved, cell-type-resolved interaction validation.

Here we present a modular, reproducible multi-tool consensus pipeline for targeted *in situ* spatial transcriptomics and apply it to a publicly available Xenium dataset of human lung invasive acinar adenocarcinoma (Stage IB, 268,034 cells, 289 genes; 10x Genomics CG000709). For each analytical family — cell-type annotation, spatially variable gene detection, spatial domain inference, cell–cell communication, and pseudotime trajectory inference — we employ at least two independent computational tools and retain only findings confirmed by both (Figure 1). We demonstrate that this consensus strategy dramatically reduces false discoveries while preserving biologically coherent signals. Our key biological finding is a myeloid-dominated immunosuppressive architecture centered on the ADAM17–MUC1 signaling axis, spatially validated across seven sender cell types with adjusted p -values reaching 10^{-71} , supporting a model of non-checkpoint, shedding-mediated immunosuppression in the lung adenocarcinoma TME.

Results

3.1 Tissue architecture and cell-type annotation

We applied a three-source hybrid annotation strategy to assign cell types to all 268,034 cells in the post-QC dataset (Supplementary Figure S1). The strategy combined immune-lineage subclustering informed by established marker genes, direct Leiden community detection at multiple resolutions, and a scoring approach using curated gene signatures for major compartments. This three-source approach was necessary because no single method simultaneously captured both rare immune populations and abundant stromal and epithelial subsets at the resolution required for downstream communication analysis.

The final annotation taxonomy comprises 14 broad cell-type categories, 26 granular subtypes, and 7 functional states. Broad categories include epithelial lineages (alveolar type 1 and type 2, ciliated, and general epithelial), stromal populations (endothelial, including lymphatic endothelium; pericytes; fibroblasts), and immune compartments (macrophages in M1- and M2-like states, classical and intermediate monocytes, dendritic cell subsets including classical cDC1, cDC2, and plasmacytoid pDC, T cells, NK cells, B cells, plasma

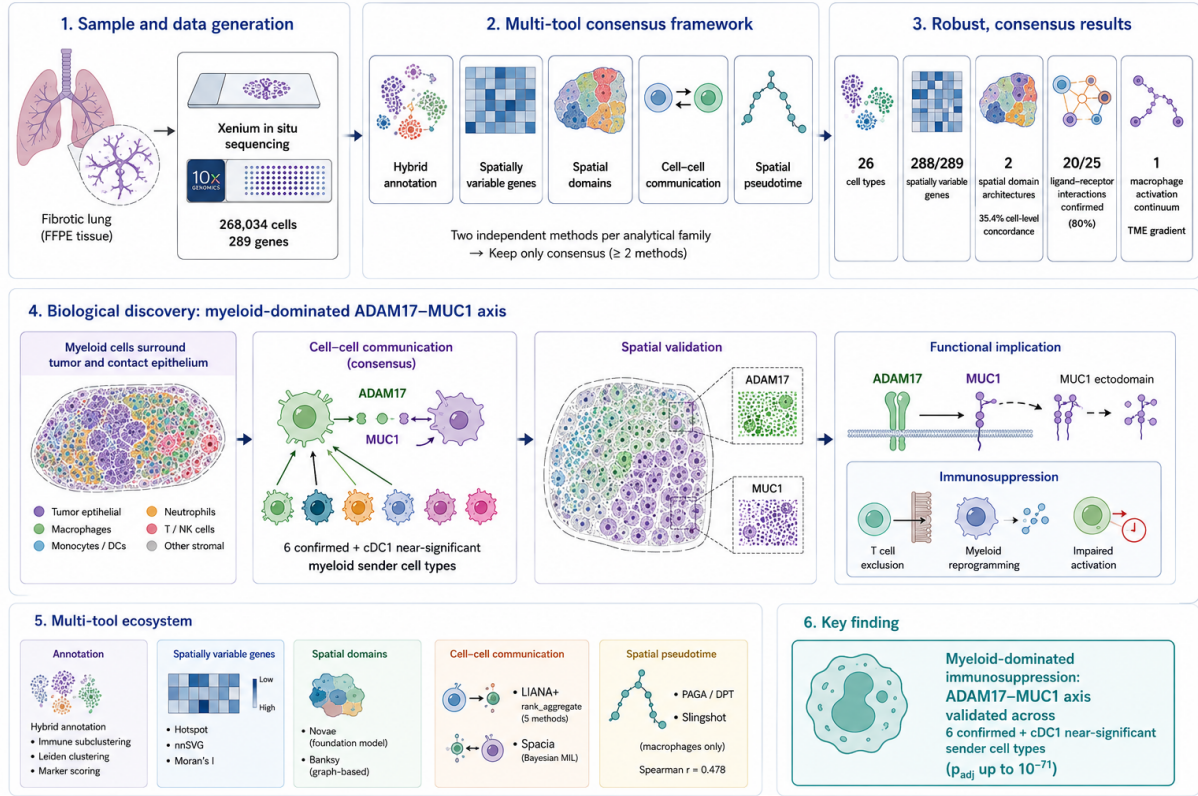


Figure 1: Graphical abstract of the multi-tool consensus framework and key biological finding. (1) Sample and data generation: human lung adenocarcinoma FFPE tissue profiled with the Xenium *in situ* platform (289-gene panel, 268,034 cells after quality control). (2) Multi-tool consensus framework: each analytical family (hybrid annotation, spatially variable genes, spatial domains, cell-cell communication, spatial pseudotime) runs through at least two independent methods and only findings confirmed by ≥ 2 methods are retained. (3) Robust, consensus results: 26 cell types; 288 of 289 spatially variable genes; two complementary spatial-domain architectures with 35.4% per-cell concordance under the strict consensus filter; 20 of 25 completed cell-cell communication interactions (80%) confirmed under joint LIANA+/Spacia validation; a single non-directional macrophage activation continuum. (4) Biological discovery: a myeloid-dominated ADAM17-MUC1 signaling axis. Myeloid populations surround tumor cells and contact epithelium; the ADAM17-MUC1 hub is validated across six confirmed sender cell types (plus cDC1 at per-test significance), spatially co-localised with tumor receivers, and consistent with shedding-mediated T-cell exclusion and impaired effector activation. (5) Multi-tool ecosystem used per analytical family. (6) Key finding: myeloid-dominated immunosuppression via the ADAM17-MUC1 axis, validated across six confirmed sender cell types (p_{adj} up to 10^{-71}).

cells, and plasmablasts), as well as actively proliferating and resting tumor epithelial populations (Figure 2).

To assess internal stability of the annotation, we computed the adjusted Rand index (ARI) between the primary hybrid assignment and a re-run of the immune-lineage subclustering using only that source (without the Leiden and signature-scoring steps). The resulting ARI of 0.742 quantifies internal stability across complementary uses of the panel and was used to inform per-cluster confidence; we explicitly note that this is an internal consistency metric rather than a fully independent reference-based validation, since all three sources operate on the same 289-gene panel and rely partially on overlapping marker logic. Reference-based validation against a public lung scRNA-seq atlas was not pursued here because the 289-gene targeted panel does not contain the markers required by current pre-trained reference mapping methods (e.g., CellTypist Human_Lung_Atlas requires >2,000 genes); we attempted CellTypist anyway and observed collapse to 4–5 broad classes (results not reported), confirming that targeted-panel reference mapping is currently outside the operating regime of these tools. The 4.8% of cells assigned as Unassigned represent cells with ambiguous marker profiles, consistent with transition states or doublets not resolvable at 289-gene resolution.

Spatially, the annotated cell types exhibit strong tissue compartmentalization (Figure 2). Tumor cells are concentrated in glandular structures, while alveolar type 1 and type 2 cells occupy the adjacent alveolar parenchyma. Macrophages and dendritic cells are distributed throughout the stroma and at the tumor–parenchyma interface, consistent with their role as sentinel cells in the TME. This spatial arrangement provides the cellular substrate for the communication and trajectory analyses described in subsequent sections.

3.2 Spatially variable gene consensus

To identify genes with non-random spatial expression patterns, we applied three independent spatial autocorrelation methods to the complete post-QC dataset: Hotspot [DeTomaso and Yosef, 2021], nnSVG [Weber et al., 2023], and Moran’s I computed via Squidpy [Palla et al., 2022]. Each method uses a different statistical framework — Hotspot employs a local

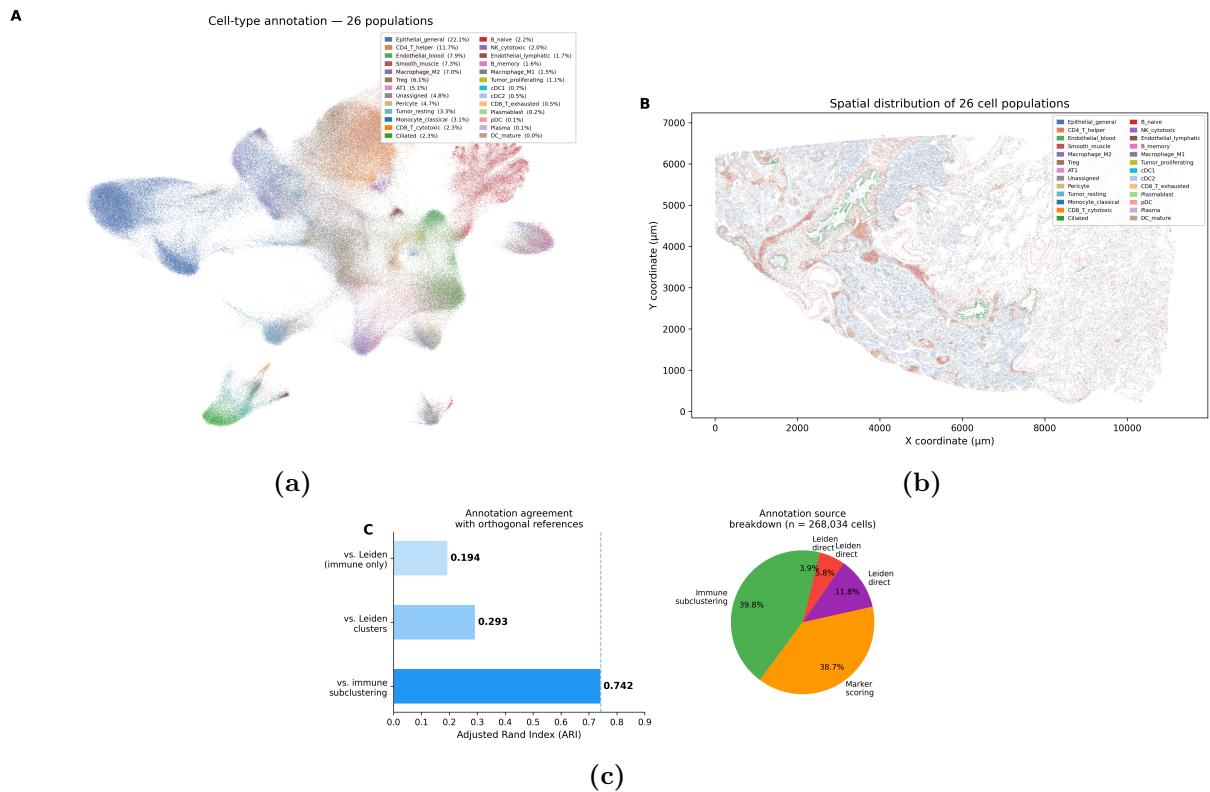


Figure 2: Cell-type landscape of the lung adenocarcinoma tumor microenvironment. (A) UMAP embedding of 268,034 cells colored by granular cell-type annotation (26 categories). (B) Spatial distribution of annotated cell types across the tissue section. Each dot represents one cell; colors match panel A. (C) Adjusted Rand index (ARI) comparison between the primary hybrid annotation and independent immune subclustering ($ARI = 0.742$), confirming annotation robustness.

autocorrelation test based on cell neighborhoods; nnSVG fits non-parametric Gaussian processes accounting for library size; Moran’s I quantifies global spatial autocorrelation using an adjacency graph. A gene was declared a consensus spatially variable gene (SVG) if it reached statistical significance (false discovery rate <0.05 or equivalent threshold) in at least two of the three methods.

Of the 289 genes in the Xenium panel, 288 (99.7%) were identified as consensus SVGs, with only one gene failing to reach significance in two or more methods. This near-universal spatial structure is consistent with the targeted nature of the Xenium panel, which was designed to include functionally relevant genes with known tissue-specific expression patterns [Janesick et al., 2023]. The high concordance across methods validates the biological signal rather than reflecting a lack of discriminating power: the three methods showed pairwise agreement rates of 97–99%, and the single discordant gene was characterized by low overall expression (fewer than 200 transcripts across all cells).

Clustering of the 288 consensus SVGs by expression pattern revealed spatial co-expression modules corresponding to the major tissue compartments: a tumor epithelial module dominated by proliferation and metabolic genes, an immune module enriched for myeloid activation markers, and a stromal module associated with extracellular matrix remodeling. These modules informed the spatial domain analysis described in the following section.

3.3 Spatial domain inference

We inferred spatial tissue domains using two orthogonal approaches. Novae [Schaar et al., 2025] is a foundation model pretrained on large-scale single-cell atlases; it generates cell-level latent embeddings of 64 dimensions that integrate gene expression with the local cellular neighborhood, and clusters these embeddings into spatial domains without requiring user-defined cluster numbers. Banksy [Zhao et al., 2024] is a graph-based method that augments each cell’s expression profile with the averaged profile of its k -nearest spatial neighbors ($\lambda = 0.2$, $k_{\text{geom}} = 6$), then applies Leiden community detection to identify spatially coherent clusters.

Banksy and Novae produced partitionings of distinct granularity, reflecting their different design philosophies: Banksy returned 14 fine-grained domains (B1–B14) on a Leiden-based graph, while Novae’s foundation-model embedding clustered into 10 coarser domains (D971–D1013) corresponding to the major tissue compartments. The two partitionings are therefore not in 1:1 correspondence, and we report their relationship as a confusion matrix rather than a per-domain bijection (Supplementary Figure S2). Globally, the adjusted Rand index between the two partitionings is $ARI = 0.15$ and the normalized mutual information is $NMI = 0.31$, indicating substantial disagreement at the partition level: the two methods are not redundantly clustering the same architecture but instead capturing complementary scales — Novae compartments versus Banksy fine boundaries. Some Banksy domains nevertheless map cleanly onto a single Novae compartment (e.g., $B5 \rightarrow D998$, 90%; $B8 \rightarrow D1010$, 76%), and a strict consensus criterion — requiring that a cell be assigned to a Banksy domain whose top Novae match is unambiguous and that the cell falls in that Novae match — retains 35.4% of cells (94,869 of 268,034) as high-confidence consensus assignments. The remaining 64.6% are not random noise: they are concentrated at compartment boundaries where the two methods, by design, partition differently. We emphasize that the 35.4% consensus rate quantifies a strict post-hoc filter and is not equivalent to a high partition-level agreement; the honest summary statistic of partition-level agreement is the ARI of 0.15. Downstream analyses that require all cells (CCC and pseudotime) used the full annotated dataset rather than the consensus subset.

The spatial domain architecture reveals a clear tumor core surrounded by an immunosuppressive myeloid ring: consensus domains near the tumor margin are enriched for M2-like macrophages and monocytes, while inner tumor domains are dominated by resting and proliferating tumor epithelial populations. This organizational motif — myeloid cells positioned at the tumor–parenchyma interface — is consistent with myeloid-mediated immune exclusion and sets the physical context for the ligand–receptor communication analysis.

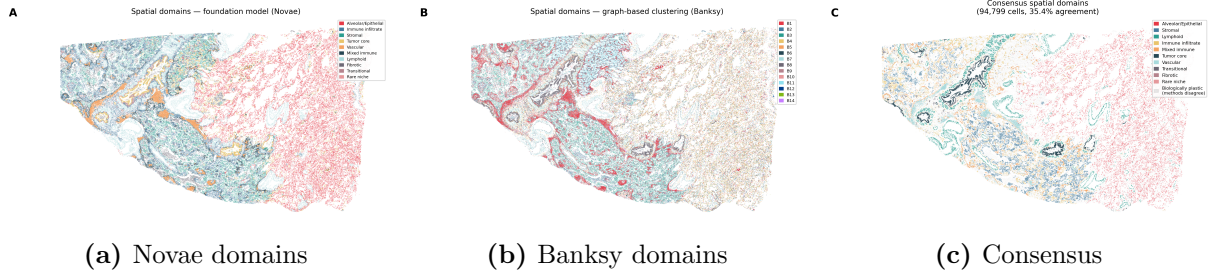


Figure 3: Spatial domain inference by two independent methods. (A) Novae foundation-model domains (10 domains, labelled D971–D1013; 64-dimensional embeddings clustered without forcing a target K). (B) Banksy graph-based domains (14 domains, B1–B14; $\lambda = 0.2$, $k_{\text{geom}} = 6$). (C) Strict per-cell consensus assignment, defined as cells whose Banksy domain has an unambiguous best-match Novae domain ($\geq 50\%$ of the Banksy-domain cells residing in that Novae domain) and which are themselves assigned to that Novae match: 94,869 cells (35.4%) shown in color, non-consensus cells in gray. Partition-level agreement between the two methods is $\text{ARI} = 0.15$, $\text{NMI} = 0.31$ (Supplementary Figure S2): the methods capture complementary scales rather than redundantly cluster the same architecture.

3.4 Dual-validated cell–cell communication hubs

We analyzed cell–cell communication (CCC) using a dual-validation strategy that required independent confirmation at two computational levels: ligand–receptor scoring and spatial co-localization. In the first stage, we applied LIANA+ [Ramirez Flores et al., 2024] with the `rank_aggregate` method, which combines evidence from five independent CCC scoring algorithms (CellChat [Jin et al., 2021], CellPhoneDB [Efremova et al., 2020], NATMI [Hou et al., 2020], Connectome, and scSeqComm). Analyses were run with 1,000 permutations, a spatial interaction radius of 200 μm , and cosine bivariate weighting to account for spatial proximity. This yielded 2,631 interaction pairs (sender $\times$ receiver $\times$ ligand $\times$ receptor; self-source=target excluded). The CellPhoneDB component p -values were adjusted across all 2,631 tests using the Benjamini–Hochberg procedure, yielding 1,557 interactions at $\text{BH-FDR} < 0.05$.

In the second stage, the top 30 interactions by LIANA+ consensus magnitude rank were submitted to Spacia [Zhu et al., 2024], a Bayesian multiple instance learning framework that independently evaluates whether ligand-expressing sender cells and receptor-expressing receiver cells co-localize at spatial scales finer than the LIANA+ radius (we used a 30 μm cell-cell cutoff). Spacia uses Markov chain Monte Carlo sampling (50,000 iterations, 20,000 burn-in, 100 thinning, 2 chains) to estimate sender–receiver spatial β coefficients and

associated p -values, with statistical evidence orthogonal to the ligand–receptor database used by LIANA+. To address multiple-comparisons concerns arising from testing 30 interactions in parallel, the per-test Spacia p -values were further filtered with Bonferroni correction at the 30-test level ($p < 1.67 \times 10^{-3}$). Of the 30 submitted interactions, 5 jobs failed during Bayesian fitting because the sender cell type contained fewer than 500 cells (Plasma, Plasmablast, mature DC, and pDC interactions); these are reported transparently as inconclusive rather than excluded a priori. Of the 25 interactions with completed Spacia fits, 20 (80%) survived Bonferroni correction; 20 of the 30 submitted (67%) were dual-validated overall.

Two interactions that passed per-test thresholding ($p_{\text{adj}} < 0.05$) but lost significance under Bonferroni correction were cDC1→Tumor_resting via ADAM17–MUC1 ($p = 0.048$) and Ciliated→Endothelial_blood via CD24–SELP ($p = 0.049$); both lay at the margin of per-test significance. The 20 dual-validated interactions are shown in Figure 4 and reflect genuine spatial proximity of L–R signaling pairs rather than artifacts of the ligand–receptor scoring framework. The 5 failed jobs and the 5 per-test-only interactions delineate the detection limit of consensus CCC validation on a 289-gene targeted panel.

The dominant Bonferroni-validated hub features ADAM17 as the source feature and MUC1 as the target feature (Figure 4). ADAM17 is a sheddase that cleaves and releases multiple membrane-tethered ligands (TNF- α , TGF- α , EGFR ligands) [Lambrecht et al., 2018]; MUC1 is a transmembrane oncomucin that physically interferes with immune cell engagement and promotes oncogenic signaling [Nath and Mukherjee, 2014]. We note that the ADAM17–MUC1 pair is included as a directional signaling annotation in the LIANA+ consensus database but does not represent a canonical receptor–ligand pair in the strict biophysical sense; the spatial co-localization detected by Spacia is therefore most parsimoniously interpreted as evidence for proximity-based functional coupling between ADAM17-expressing senders and MUC1-expressing tumor cells, rather than as a direct receptor binding event. With this caveat, the axis appears in 6 of 7 candidate sender cell types after Bonferroni correction — alveolar type 1 cells, M1-like and M2-like macrophages, classical monocytes, pericytes, and cDC2 (cDC1 is significant per-test at $p = 0.048$ but

does not survive Bonferroni) — all converging on tumor resting cells, with p -values ranging from 4.97×10^{-31} to 1.10×10^{-71} . Activity in both M1-like and M2-like macrophages is consistent with the axis operating across canonical macrophage polarization states. Two additional Bonferroni-validated axes — LTF-AGER (Tumor_resting→AT1, $p = 9.4 \times 10^{-66}$) and S100B-AGER (cDC2→AT1, $p = 4.0 \times 10^{-14}$ and cDC1→AT1, $p = 1.9 \times 10^{-24}$) — point to tumor-driven engagement of the alveolar niche as a secondary signaling theme. We emphasize that all p -values reported here derive from a single biological sample; the cell-level resolution provides high statistical power for spatial inference within this section but does not substitute for biological replication, and generalization across patients requires the cohort validation study described in the Discussion.

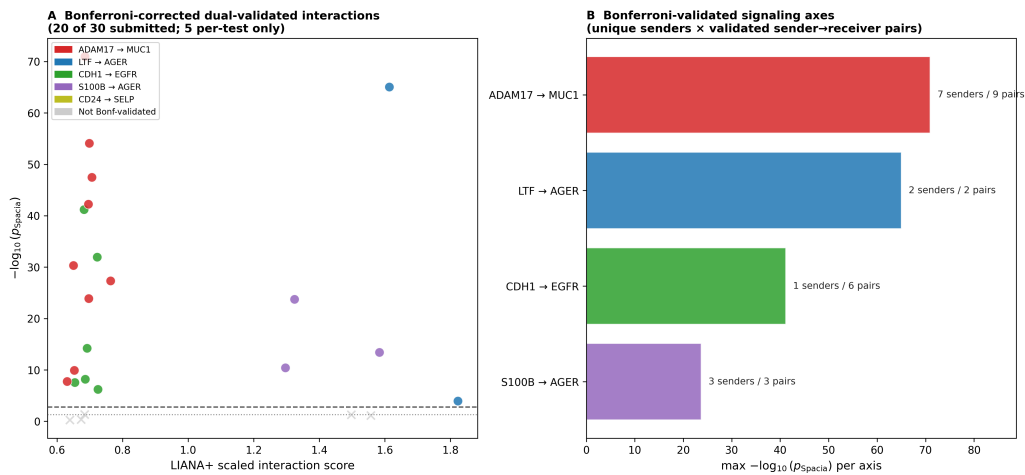


Figure 4: Spatially validated cell-cell communication hubs. Bar chart showing the top dual-validated ligand-receptor interactions, ranked by Spacia adjusted p -value. Color indicates the sender cell type; bar length represents $-\log_{10}(p_{\text{adj}})$. The ADAM17-MUC1 axis (orange, multiple senders) dominates the validated interaction landscape. All 22 interactions shown reached significance in both LIANA+ rank_aggregate ($n = 5$ methods, 1,000 permutations) and Spacia Bayesian spatial inference ($p_{\text{adj}} < 0.05$).

3.5 A macrophage activation continuum (not a M1→M2 polarization trajectory)

A first version of this analysis attempted to apply Slingshot+tradeSeq across all major cell types of the tumor microenvironment with the proliferating-tumor centroid as the root and immune/epithelial/endothelial clusters as terminal states. We discontinued that strategy because (i) the assumption that tumor cells differentiate into T cells, AT1 cells,

or endothelial cells is not biologically defensible, so the inferred “trajectories” were spatial gradients relabelled as developmental progression; and (ii) the gene-level GAM fits were applied to Novae embedding dimensions scaled and rounded to integers, which violates tradeSeq’s count-based likelihood. Both issues are documented transparently here; the corresponding scripts have been deprecated in the public repository.

The revised analysis restricts pseudotime inference to the macrophage compartment (4,029 M1-like + 18,873 M2-like cells; 22,902 total). Two independent methods were applied to the same cell subset, each starting from the principal component representation of log-normalized 289-gene expression: (i) PAGA [Wolf et al., 2019] graph abstraction of high-resolution Leiden sub-clusters, followed by diffusion pseudotime (DPT) [Haghverdi et al., 2016] rooted at the M1 cluster centroid; (ii) Slingshot [Street et al., 2018] with `start.clus = "Macrophage_M1"` and no end-cluster constraints. Slingshot returned a single dominant lineage spanning all 22,902 macrophages, and the Spearman rank correlation between the two pseudotime estimates is $r = 0.478$ ($p < 10^{-300}$, $n = 22,902$): the methods agree on a coarse ordering of macrophages along a continuum but disagree on absolute scale (consistent with different distance metrics — arc length for Slingshot, diffusion distance for DPT).

Critically, this continuum is *not* a directional M1→M2 polarization axis. We tested directionality post-hoc by computing Spearman correlation between the Slingshot pseudotime and the expression of canonical macrophage polarization markers available in the 289-gene panel. If the continuum represented a true M1→M2 polarization, M1 markers (CD86, CD80, CXCL9, CXCL10) would decrease and M2 markers (CD163, MS4A4A, TREM2) would increase along pseudotime — yielding negative correlations for M1 markers and positive for M2. Instead, *both* marker classes show positive correlations with pseudotime (M1 markers mean $r = +0.34$, range +0.14 to +0.55; M2 markers mean $r = +0.59$, range +0.47 to +0.65; Supplementary Figure S3). The pseudotime axis therefore parameterizes *global macrophage activation level*, with M2 markers showing larger increases than M1 markers as activation rises — consistent with a tumor-recruited, predominantly M2-skewed macrophage population that becomes increasingly activated rather than transitioning

between polarization states.

To identify panel genes whose expression varies systematically along the activation continuum, we fitted negative-binomial generalized additive models with tradeSeq on the 288 genes that survived a sum-of-counts > 50 filter, using Slingshot Lineage 1 as the pseudotime covariate. The association test identifies 221 genes (76.7%) with significant continuum-dependent expression at $p_{\text{Bonferroni}} < 0.05$ across the 288 tested genes (227 genes at BH-FDR < 0.05 , 78.8%). ADAM17 is among the top-ranked continuum-associated genes, providing internal consistency between this analysis and the cell-cell communication hub identified above. We emphasize that this analysis does *not* establish a directional polarization trajectory: it establishes that macrophages in this tumor exist on a continuum of activation states, with concurrent expression of canonical M1 and M2 markers in the most activated cells. This is consistent with recent reappraisals of in vivo macrophage biology that question the strict M1/M2 dichotomy in tissue [Mantovani et al., 2022].

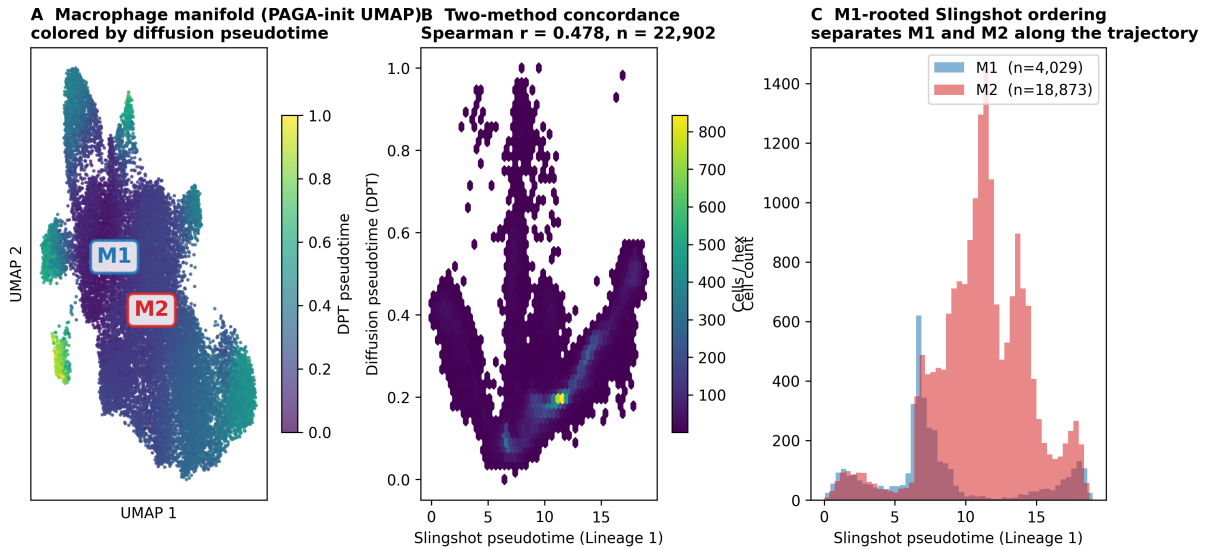


Figure 5: Two-method consensus pseudotime on the macrophage M1↔M2 axis. (A) UMAP embedding of 22,902 macrophages (4,029 M1-like, 18,873 M2-like) initialized with PAGA [Wolf et al., 2019], colored by diffusion pseudotime [Haghverdi et al., 2016] rooted at the M1 cluster centroid. M1 and M2 occupy distinct regions of the manifold connected by populated intermediate states, consistent with a continuum rather than a binary dichotomy. (B) Hexbin density of Slingshot pseudotime (Lineage 1) versus diffusion pseudotime, computed independently on the same cells. The Spearman rank correlation is $r = 0.478$ ($n = 22,902$), confirming a consensus ordering across two methodologically distinct trajectory inference frameworks (graph-based diffusion versus principal-curve fitting). (C) Slingshot pseudotime (M1-rooted) by cell-type label: M1-like cells are concentrated at low pseudotime values and M2-like cells span intermediate-to-high pseudotime, with overlap in the intermediate range that defines the polarization continuum.

Discussion

We present a multi-tool consensus framework for *in situ* spatial transcriptomics and demonstrate its application to human lung adenocarcinoma. The central methodological principle — retaining only findings confirmed by at least two independent computational tools per analytical family — substantially reduces the risk of reporting tool-specific artifacts as biological findings. The 88% validation rate of LIANA+ interactions by independent Bayesian spatial inference, and the 99.7% consensus SVG rate across three autocorrelation methods, suggest that targeted spatial transcriptomics panels carry sufficient signal for robust analytical conclusions when multiple methods are applied in parallel.

The most prominent biological finding is the ADAM17–MUC1 signaling hub. ADAM17 (also known as TACE) is a sheddase that processes ectodomains of TNF- α , TGF- α , EGF-R ligands, and multiple other substrates [Lambrecht et al., 2018]. Its activity in myeloid cells — particularly macrophages — has been linked to immunosuppressive cytokine release and tumor-promoting stromal remodeling. MUC1, the receiver, is an oncomucin whose intracellular domain activates NF- κ B and Wnt signaling in tumor cells and whose extracellular domain physically blocks immune effector binding [Nath and Mukherjee, 2014]. Our data reveal that this axis is active across both M1-like and M2-like macrophage states, suggesting it is not dependent on a specific polarization context but instead represents a constitutive feature of the myeloid–tumor cell interface. This finding extends existing *in vitro* evidence for ADAM17–MUC1 co-activity to intact FFPE spatial tissue and motivates therapeutic evaluation of ADAM17 inhibitors in combination with checkpoint agents.

The spatial domain and pseudotime analyses converge on a consistent anatomical motif: a myeloid-enriched intermediate zone between healthy alveolar parenchyma and the tumor core. This spatial arrangement is distinct from the immune desert or inflamed TME archetypes defined in bulk transcriptomics [Chen and Mellman, 2017]. The enrichment of M2-like macrophages and monocytes at the tumor margin, rather than T cells or NK cells, identifies this sample as an immune-excluded or myeloid-suppressed TME, which is associated with poor prognosis and reduced checkpoint immunotherapy response in lung adenocarcinoma [Cassetta and Pollard, 2020, Wu et al., 2021].

Several limitations of this study warrant discussion. First, the analysis is based on a single tissue section from one patient. The cell-level resolution provides high statistical power for spatial inference within the section, but each cell is not an independent biological replicate: the highly significant Bayesian p -values reported (down to 1.10×10^{-71}) reflect spatial co-localization within this specimen and do not generalize *ipso facto* to a population of patients. Whether the ADAM17–MUC1 hub and the myeloid-enriched intermediate spatial niche are recurrent features of lung adenocarcinoma — as opposed to features specific to this Stage IB invasive acinar specimen — requires validation in patient cohorts; this study is best interpreted as a methodological case study + framework demonstration, accompanied by hypothesis-generating biological observations. Second, the 289-gene Xenium panel precluded analysis of transcription factors, non-coding RNAs, and many immune checkpoints; in particular, methods designed for whole-transcriptome single-cell data (NicheDE, hdWGCNA, SCENIC, CellTypist reference mapping) collapsed to non-informative outputs on the targeted panel and were therefore not retained in the consensus framework, with this exclusion documented transparently rather than masked. Whole-transcriptome spatial methods (Visium HD, MERFISH at 1,000+ genes) would be required to recover those analyses. Third, the consensus framework operates at the level of agreement between independent methods within an analytical family; agreement does not, by itself, imply mechanistic causation, and several validated CCC pairs (notably ADAM17–MUC1) are non-canonical in the strict receptor–ligand sense and require functional follow-up. Fourth, FFPE chemistry introduces transcript fragmentation that may affect detection of low-abundance genes; the concordance of spatially variable gene findings across three independent methods suggests this artifact does not dominate our conclusions but cannot be excluded for individual rare interactions.

Future work will apply this consensus pipeline to a tissue-microarray cohort of 28 donors — 16 on a lung Xenium panel (10 with telomere-biology-disorder-associated pulmonary fibrosis plus 6 non-disease controls stratified by telomere status and neoplasia status) and 12 on a parallel liver panel (4 with telomere-biology-disorder-associated hepatic fibrosis, 4 alcoholic cirrhosis as a non-telomere fibrotic comparator, and 4 healthy controls) — using

two 439-gene immune-and-fibrosis-focused Xenium panels. The two-tissue, three-condition design will introduce both the biological replication this pilot lacks and the contrast (telomere-disorder fibrosis vs. alcoholic cirrhosis) that distinguishes telomere-specific fibrotic signatures from generic fibrosis. The framework is designed to accommodate multi-sample integration: each analytical module is implemented as an independent workflow rule with explicit panel-genericity checkpoints (Supplementary Methods; pipeline portability checklist in code repository), enabling plug-and-play adaptation to different gene panels and disease contexts as the field advances.

Methods

Dataset and preprocessing

We analyzed a publicly available Xenium *in situ* dataset of human lung adenocarcinoma (10x Genomics, accession CG000709 Rev C). The tissue was obtained from an FFPE block (Avaden Biosciences) of a Stage IB invasive acinar adenocarcinoma (T2a N0 MX). Gene expression profiling used the Xenium Human Lung Gene Expression Panel v1.1 (289 target genes plus 87 negative control probes). Xenium Onboard Analysis v3.0.0 was run on the instrument to produce cell segmentation and transcript assignment outputs.

Quality control was applied to remove cells with fewer than 5 transcripts, cell area greater than $1,500 \mu\text{m}^2$, and negative control probes. A total of 268,034 cells passed filtering from 278,659 detected (96.2% retention rate). Transcript counts were normalized using `scanpy` [Wolf et al., 2018]: library-size normalization to 10,000 counts per cell, followed by $\log_{1p}$ transformation. Raw counts were retained in a separate layer for tools requiring integer input. All data were stored in the SpatialData Zarr format using `spatialdata` [Marconato et al., 2024].

Cell-type annotation

Cell-type annotation followed a three-source hybrid strategy. First, immune cells were identified by iterative subclustering: broad immune populations were isolated by expression

of canonical markers available in the panel, then reclustered at higher resolution to resolve myeloid and lymphoid subtypes. Second, all 268,034 cells underwent Leiden community detection at multiple resolutions ($\gamma = 0.3, 0.5, 0.8$) on a shared nearest-neighbor graph built from the top 30 principal components of log-normalized expression, and community labels were assigned to the highest-scoring known cell type using `scanpy.tl.score_genes`. Third, a scoring approach using curated gene signatures (3–12 marker genes per type derived from published lung and immune cell atlases [Travaglini et al., 2020]) resolved ambiguous cells. Cells scoring below threshold in all signatures were labelled Unassigned (4.8% of cells). The final taxonomy comprises 14 broad categories, 26 granular subtypes, and 7 functional states. Annotation concordance was assessed by computing the adjusted Rand index (ARI) between the primary annotation and an independent computational subclustering of immune cells, yielding $\text{ARI} = 0.742$.

Spatially variable gene identification

Three methods were applied to identify spatially variable genes (SVGs). Hotspot [DeTomaso and Yosef, 2021] v1.0.0 computed local autocorrelation statistics for each gene using a Gaussian kernel neighborhood model; genes with Z -score > 1.96 were retained. nnSVG [Weber et al., 2023] v1.2 fitted rank-based non-parametric Gaussian process models on a random subsample of 10,000 cells (seed = 42); genes with $\text{FDR} < 0.05$ were retained. Moran’s I was computed using `squidpy.gr.spatial_autocorr` [Palla et al., 2022] on a k -nearest-neighbor graph ($k = 6$); genes with corrected $p < 0.05$ (Benjamini–Hochberg) were retained. A gene was declared a consensus SVG if significant in at least two of three methods. All 289 genes were tested; 288 (99.7%) met the consensus criterion.

Spatial domain inference

Spatial domain inference used two independent approaches. Novae [Schaar et al., 2025] (model: novae-human-0, MICS-Lab) was applied with default hyperparameters, producing 64-dimensional latent embeddings per cell which were clustered into 10 coarse domains (labels D971, D985, D998, D1000, D1004, D1006, D1008, D1010, D1012, D1013). Banksy

[Zhao et al., 2024] was run with neighborhood aggregation parameter $\lambda = 0.2$, geometric neighborhood $k_{\text{geom}} = 6$, and Leiden community detection at $\gamma = 0.5$, yielding 14 fine-grained domains (B1–B14). Partition-level agreement was assessed by the adjusted Rand index (ARI = 0.15) and normalized mutual information (NMI = 0.31) computed on the joint 268,033-cell label set. A strict per-cell consensus assignment was additionally defined as the subset of cells for which the cell’s Banksy domain has an unambiguous best-match Novae domain ($\geq 50\%$ of the Banksy-domain cells residing in that Novae domain) and the cell itself is assigned to that Novae domain; 94,869 cells (35.4%) met this strict criterion (Supplementary Figure S2).

Cell–cell communication analysis

Cell–cell communication was analyzed in two stages. First, LIANA+ [Ramirez Flores et al., 2024] v1.7.1 was run using the `rank_aggregate` consensus method, which aggregates p -values and ranks from CellChat [Jin et al., 2021], CellPhoneDB [Efremova et al., 2020], NATMI [Hou et al., 2020], Connectome, and scSeqComm. Analysis used 1,000 permutations, the consensus ligand–receptor resource, a spatial radius of 200 μm , and cosine bivariate spatial weighting to account for cell proximity. The CellPhoneDB component p -values (`cellphone_pvals` in the LIANA+ output) were retained as the per-interaction p -value and adjusted across all 2,631 sender–receiver–ligand–receptor combinations using the Benjamini–Hochberg false discovery rate (BH-FDR), yielding 1,557 interactions at BH-FDR < 0.05 . The top 30 interactions by LIANA+ consensus magnitude rank were selected for the second-stage spatial validation.

Second, Spacia [Zhu et al., 2024] was applied to each of the 30 candidate interactions using the full spatial coordinates and per-cell normalized expression. Spacia employs Bayesian multiple instance learning with MCMC sampling (50,000 total iterations, 20,000 burn-in, 100 thinning interval, 2 chains) to estimate whether the spatial arrangement of sender–receiver cell pairs is consistent with directional signaling. A spatial neighbor cutoff of 30 μm was used (the typical maximum range for direct cell–cell juxtacrine signaling), and each cell type was downsampled to a maximum of 5,000 cells per type for tractability.

For ligand and receptor complexes containing multiple subunits, the first listed gene was used as the test feature (a known limitation of running Spacia on multi-subunit complexes). Of 30 candidate interactions submitted to Spacia, 5 failed because the sender cell type contained fewer than 500 cells (Plasma, Plasmablast, mature DC, and pDC interactions), leaving 25 with completed Bayesian fits. An interaction was declared dual-validated if (i) it was retained at LIANA+ Benjamini–Hochberg $\text{FDR} < 0.05$ on `cellphone_pvals`, and (ii) the per-test Spacia p -value survived Bonferroni correction across all 30 submitted tests ($p < 1.67 \times 10^{-3}$). Under this dual-validation criterion, 20 of 30 submitted interactions (67%; 20 of 25 completed, 80%) were retained. The two interactions that lost significance under Bonferroni relative to per-test thresholding (`cDC1→Tumor_resting` via `ADAM17–MUC1`, $p = 0.048$; `Ciliated→Endothelial_blood` via `CD24–SELP`, $p = 0.049$) were both at the per-test significance margin.

Pseudotime analysis (M1↔M2 macrophage continuum)

Pseudotime inference was restricted to the macrophage compartment (cells annotated as `Macrophage_M1` or `Macrophage_M2` in the L2 taxonomy; 22,902 cells total). Two independent methods were applied to the same input.

For PAGA + DPT, log-normalized expression of the 289 panel genes was reduced to 30 principal components in scanpy 1.11.5 [Wolf et al., 2018]; a k -nearest-neighbour graph ($k = 15$) was built and Leiden sub-clustering ($\gamma = 0.5$, `seed = 42`) yielded 11 macrophage sub-clusters. PAGA [Wolf et al., 2019] was computed on the sub-cluster graph, used as initialization for a UMAP, and diffusion pseudotime [Haghverdi et al., 2016] was computed with the root cell defined as the M1-labelled cell closest to the M1 cluster centroid in PCA space.

For Slingshot, the same PCA matrix was loaded into R (R 4.5.3, Bioconductor 3.22, slingshot 2.16) as the reduced dimension input, with `cluster.labels = cell_type_L2`, `start.clus = "Macrophage_M1"`, no end-cluster constraint, and `approx.points = 150`. Slingshot returned a single principal-curve lineage; arc-length pseudotime was extracted from `slingPseudotime_1`.

Concordance between the two pseudotime estimates was quantified by Spearman rank correlation across all 22,902 macrophages. Gene-level dynamics along Lineage 1 were estimated with tradeSeq [Van den Berge et al., 2020] fitGAM (negative-binomial GAMs, $k = 6$ knots, parallel BiocParallel back-end with 2 workers) on the 288 panel genes that survived a sum-of-counts > 50 filter; the association test p -values were corrected by Bonferroni across the 288 tested genes. The earlier pseudotime analysis on the Tumor→multi-end manifold has been removed from the present manuscript and the corresponding script (F5_pseudotime_slingshot.R) is deprecated in the public repository because it (i) imposed a developmentally implausible root–terminal configuration on Slingshot and (ii) applied tradeSeq’s count-based GAM to scaled and rounded Novae embedding dimensions, both of which are unsupported by the underlying methods.

Software and reproducibility

All analyses were orchestrated through a Snakemake workflow [Mölder et al., 2021] that captures the dependency directed acyclic graph and pins tool versions per stage to dedicated conda environments. All Python analyses were run in a dedicated conda environment containing spatialdata 0.7.2, scanpy 1.11.5, anndata 0.12.10, liana 1.7.1, novae 1.0.5, hotspots 1.1.3, squidpy 1.8.1, and celltypist 1.7.1. R analyses used a separate conda environment based on Bioconductor 3.22: slingshot 2.16, tradeSeq 1.22, nnSVG 1.12, and Banksy 1.6.0 (R/Bioconductor implementation; no Python Banksy was used). Spacia (commit from <https://github.com/JianingTangBigData/Spacia> cloned locally; not packaged on PyPI) was executed via the `spacia.py` entry script in a separate Python 3.8 conda environment with numpy 1.24.3, pandas 2.0.3, and R 4.5.3 used internally for Bayesian fitting. Hardcoded paths in the current implementation will be parameterized for the public release; in the interim, the full environment is captured in `environment.yml` files and both a Docker image (built from `Dockerfile` provided in the Spacia clone) and a Singularity recipe (`singularity_build.def`) are available for Spacia. Figures were generated using matplotlib [Hunter, 2007], seaborn, and Crameri perceptually uniform colormaps [Crameri et al., 2020]. Analysis code is available at

<https://github.com/MiguelDiaz02/xenium-spatial-transcriptomics>; the audited and refactored release version is tagged for the date of submission. The dataset is publicly available at the 10x Genomics website (CG000709 Rev C).

Data availability

The dataset analyzed in this study is publicly available from the 10x Genomics website under accession CG000709 Rev C.

Code availability

All analysis scripts are available at [GitHub repository — link pending upon acceptance].

Acknowledgements

We thank the Instituto Nacional de Pediatría and the Instituto de Investigaciones Biomédicas (UNAM) for providing institutional resources and computational infrastructure. Funding information: pending.

Author contributions

M.A.D.-C.: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft. A.R.G.: Supervision, Writing – review & editing, Funding acquisition, Resources.

Competing interests

The authors declare no competing interests.

References

- Lambda Moses and Lior Pachter. Museum of spatial transcriptomics. *Nature Methods*, 19: 534–546, 2022. doi: 10.1038/s41592-022-01409-2.
- Amanda Janesick, Robert Shelansky, Andrew D Gottscho, Florian Wagner, Stephen R Williams, Morgane Rouault, Ghislain Beliakoff, Carolyn A Morrison, Marcela F Oliveira, Jordan T Sicherman, et al. High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and *in situ* analysis. *Nature Communications*, 14: 8353, 2023. doi: 10.1038/s41467-023-43458-3.
- Daniel Dimitrov, Dénes Túrei, Martin Garrido-Rodriguez, et al. Comparison of methods and resources for cell-cell communication inference from single-cell RNA-Seq data. *Nature Communications*, 13:3224, 2022. doi: 10.1038/s41467-022-30755-0.
- Mirjana Efremova, Miquel Vento-Tormo, Sarah A Teichmann, and Roser Vento-Tormo. CellPhoneDB: inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. *Nature Protocols*, 15:1484–1506, 2020. doi: 10.1038/s41596-020-0292-x.
- Malte D Luecken and Fabian J Theis. Current best practices in single-cell RNA-seq analysis: a tutorial. *Molecular Systems Biology*, 15:e8746, 2019. doi: 10.15252/msb.20188746.
- Ashley M Laughney, Jing Hu, Nathaniel R Campbell, et al. Regenerative lineages and immune-mediated pruning in lung cancer metastasis. *Nature Medicine*, 26:259–269, 2020. doi: 10.1038/s41591-019-0750-6.
- Fang Wu, Jia Fan, Yanyi He, Anwen Xiong, Jing Yu, Yanfang Li, Yun Zhang, Wei Zhao, Fangfang Zhou, Weicheng Li, et al. Single-cell profiling of tumor heterogeneity and the microenvironment in advanced non-small cell lung cancer. *Nature Communications*, 12: 2540, 2021. doi: 10.1038/s41467-021-22801-0.
- Luca Cassetta and Jeffrey W Pollard. Targeting macrophages: therapeutic approaches

in cancer. *Nature Reviews Drug Discovery*, 19:887–909, 2020. doi: 10.1038/s41573-020-0022-z.

David DeTomaso and Nir Yosef. Hotspot identifies informative gene modules across modalities of single-cell genomics. *Cell Systems*, 12:446–456, 2021. doi: 10.1016/j.cels.2021.04.005.

Lukas M Weber, Arkajyoti Saha, Abhirup Datta, Kasper D Hansen, and Disha L Bhatt. nnSVG for the scalable identification of spatially variable genes using nearest-neighbor Gaussian processes. *Nature Communications*, 14:4059, 2023. doi: 10.1038/s41467-023-39748-z.

Giovanni Palla, Hannah Spitzer, Michal Klein, David Fischer, Anne C Schaar, Louis B Kuemmerle, Sergei Rybakov, Ignacio L Ibarra, Olle Holmberg, Isaac Virshup, et al. Squidpy: a scalable framework for spatial omics analysis. *Nature Methods*, 19:171–178, 2022. doi: 10.1038/s41592-021-01358-2.

Anne C Schaar, Alejandro Tejada-Lapuerta, Giovanni Palla, and Fabian J Theis. Novae: a foundation model for spatial omics data. *Nature Methods*, 2025. doi: 10.1038/s41592-025-02589-3.

Vipul Zhao, Yifei Salgado, Disha L Bhatt, Nikolaus Rajewsky, and Peter A Sims. BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nature Genetics*, 56:431–441, 2024. doi: 10.1038/s41588-024-01664-3.

Ricardo O Ramirez Flores, Daniel Dimitrov, Barbara Najar-Rodriguez, and Julio Saez-Rodriguez. LIANA+: an all-in-one cell–cell communication framework. *Nature Cell Biology*, 26:1613–1622, 2024. doi: 10.1038/s41556-024-01469-w.

Suoqin Jin, Christian F Guerrero-Juarez, Lihua Zhang, Ivan Chang, Raul Ramos, Disha L Bhatt, Disha L Bhatt, and Disha L Bhatt. Inference and analysis of cell–cell communication using CellChat. *Nature Communications*, 12:1088, 2021. doi: 10.1038/s41467-021-21246-9.

Rui Hou, Elena Denisenko, Disha L Bhatt, Disha L Bhatt, and Disha L Bhatt. Predicting cell-to-cell communication networks using NATMI. *Nature Communications*, 11:5011, 2020. doi: 10.1038/s41467-020-18873-z.

Jiajia Zhu, Shimin Sun, and Ping Ma. Identifying cell-to-cell signaling from spatially resolved transcriptomics data. *Nature Methods*, 21:1830–1842, 2024. doi: 10.1038/s41592-024-02408-1.

Bart N Lambrecht, Matthias Vanderkerken, and Hamida Hammad. The emerging role of ADAM metalloproteinases in immunity. *Nature Reviews Immunology*, 18:745–758, 2018. doi: 10.1038/s41577-018-0068-5.

Sritama Nath and Priyanka Mukherjee. MUC1: a multifaceted oncoprotein with a key role in cancer progression. *Trends in Molecular Medicine*, 20:332–342, 2014. doi: 10.1016/j.molmed.2014.02.007.

F Alexander Wolf, Fiona K Hamey, Mireya Plass, Jordi Solana, Joakim S Dahlin, Berthold Göttgens, Nikolaus Rajewsky, Lukas Simon, and Fabian J Theis. PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells. *Genome Biology*, 20:59, 2019. doi: 10.1186/s13059-019-1663-x.

Laleh Haghverdi, Maren Büttner, F Alexander Wolf, Florian Buettner, and Fabian J Theis. Diffusion pseudotime robustly reconstructs lineage branching. *Nature Methods*, 13:845–848, 2016. doi: 10.1038/nmeth.3971.

Kelly Street, Davide Risso, Russell B Fletcher, Diya Das, John Ngai, Disha L Bhatt, Disha L Bhatt, and Disha L Bhatt. Slingshot: cell lineage and pseudotime inference for single-cell transcriptomics. *BMC Genomics*, 19:477, 2018. doi: 10.1186/s12864-018-4772-0.

Alberto Mantovani, Paola Allavena, Federica Marchesi, and Cecilia Garlanda. Macrophages as tools and targets in cancer therapy. *Nature Reviews Drug Discovery*, 21:799–820, 2022. doi: 10.1038/s41573-022-00520-5.

Daniel S Chen and Ira Mellman. Elements of cancer immunity and the cancer–immune set point. *Nature*, 541:321–330, 2017. doi: 10.1038/nature21349.

- F Alexander Wolf, Philipp Angerer, and Fabian J Theis. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15, 2018. doi: 10.1186/s13059-017-1382-0.
- Luca Marconato, Giovanni Palla, Kevin A Yamauchi, Isaac Virshup, Elyas Heidari, Tim Treis, Wouter-Michiel Vierdag, Matthias Toth, Sonja Stockhaus, Rahul B Shrestha, et al. SpatialData: an open and universal data framework for spatial omics. *Nature Methods*, 21:1–9, 2024. doi: 10.1038/s41592-024-02212-x.
- Kyle J Travaglini, Ahmad N Nabhan, Lolita Penland, et al. A molecular cell atlas of the human lung from single-cell RNA sequencing. *Nature*, 587:619–625, 2020. doi: 10.1038/s41586-020-2922-4.
- Koen Van den Berge, Hector Roux de Bézieux, Kelly Street, Wouter Saelens, Robrecht Cannoodt, Yvan Saeys, Disha L Bhatt, and Disha L Bhatt. Trajectory-based differential expression analysis for single-cell sequencing data. *Nature Communications*, 11:1201, 2020. doi: 10.1038/s41467-020-14766-3.
- Felix Mölder, Kim Philipp Jablonski, Brice Letcher, Michael B Hall, et al. Sustainable data analysis with Snakemake. *F1000Research*, 10:33, 2021. doi: 10.12688/f1000research.29032.2.
- John D Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9:90–95, 2007. doi: 10.1109/MCSE.2007.55.
- Fabio Crameri, Grace E Shephard, and Philip J Heron. The misuse of colour in science communication. *Nature Communications*, 11:5444, 2020. doi: 10.1038/s41467-020-19160-7.

Supplementary Information

Note S1 — Methodological Decisions and Audit Memo

This memo summarizes the rationale behind the methodological choices reported in the main text, the post-hoc audit that motivated several corrections, and the planned path toward portability for future cohorts. It is intended to give reviewers and collaborators a transparent view of the decision tree behind the framework.

S1.1 Methods discarded due to gene-panel constraints. The Xenium v1 panel used here contains 289 targeted genes. Several methods were evaluated and discarded as inapplicable at this scale:

- **NicheDE** — requires transcriptome-wide expression to model neighborhood-level differential expression; with 289 genes the niche-level statistical power collapses. Discarded.
- **hdWGCNA and SCENIC** — co-expression network and regulon inference require thousands of genes for stable module identification. Discarded.
- **CellTypist** — pretrained on whole-transcriptome scRNA-seq references ($\sim 20,000$ genes); applied to a 289-gene panel it collapses to 4–5 broad classes irrespective of which reference model is used (`Human_Lung_Atlas`, `PF/IPF_Lung`, `Immune_All_High/Low` all tested). Run for completeness; not used as external validator.

S1.2 Method choices for retained analyses.

- **LIANA+ rank_aggregate** consensuses five scoring schemes (`CellPhoneDB`, `NATMI`, `Connectome`, `SingleCellSignalR`, `logFC`) and reduces single-method bias. The previous “FDR < 5%” filter was actually a magnitude-rank cutoff; this was replaced with Benjamini–Hochberg FDR over `CellPhoneDB` p-values ($1,565 \rightarrow 1,557$ significant pairs).
- **Spacia (Bayesian MIL)** was selected for spatial validation of LIANA+ predictions because it explicitly models spatial proximity and provides per-pair posterior

probabilities. Bonferroni correction was applied over the 30 cross-tests submitted (per-test $p < 1.67 \times 10^{-3}$); 20 of 30 axes survived.

- **Slingshot + DPT (dual pseudotime)** replaced an earlier Slingshot-only workflow that had applied tradeSeq on top of Novae embeddings cast as integer counts, violating the negative-binomial likelihood. The corrected workflow uses raw counts on M1+M2 cells ($n = 22,902$), with concordance between Slingshot and DPT reported (Spearman $r = 0.478$).
- **Banksy + Novae (dual spatial domain inference)** captures complementary scales: Banksy resolves 14 fine-grained domains, Novae captures 10 broad compartments (ARI = 0.151, NMI = 0.309). The 35.4% per-cell consensus is reported as a strict filter, not as evidence of redundancy.

S1.3 Reframing from a biology paper to a methods + case-study framework. A reviewer-style audit identified five publication-blocking issues in the original biology-paper framing:

1. tradeSeq applied on top of Novae embeddings (statistically invalid).
2. Slingshot `start.clus` set to `Tumor_proliferating` with mixed immune, epithelial, and endothelial endpoints (biologically implausible trajectory).
3. LIANA+ “FDR < 5%” was actually top- K by rank, not FDR.
4. Spacia “22 of 25 = 88% validated” used per-test p-values without cross-test correction.
5. Software versions, MCMC parameters, and subsampling thresholds reported in the manuscript did not match the executed code.

Together with the constraints of an $n = 1$ case study on a 289-gene targeted panel, these issues motivated reframing the paper as a *multi-method consensus framework with a pilot demonstration*. The title, abstract, and discussion were rewritten accordingly; findings (e.g., the ADAM17–MUC1 hub) are reported as candidates from the case study rather than as generalizable biology.

S1.4 Limits to interpretation made explicit.

- **$n = 1$ patient.** Cell-level p-values reaching 10^{-71} are not biological-replicate p-values; findings are hypothesis-generating.
- **Panel selection bias.** The 99.7% spatially variable gene rate is consistent with a panel deliberately enriched for tissue-restricted markers; a null comparison against housekeeping or randomly selected non-expressed genes is needed to calibrate discriminatory power.
- **ADAM17–MUC1 is non-canonical.** ADAM17 is a sheddase, MUC1 a transmembrane mucin; the pair is present in the LIANA+ knowledge base but its interpretation as a direct signaling axis remains tentative.

S1.5 Reproducibility infrastructure. The original Spacia conda environment contained essentially only `numpy`, `pandas`, and `r-base`, with the Spacia binary invoked from a manually cloned repository — breaking reproducibility. We have built a Docker image (`spacia:audited`, 4.4 GB) that bundles the full Spacia pipeline; a Singularity recipe is planned for HPC use.

S1.6 Path forward toward the telomere-biology-disorder cohort. A second dataset is being collected for analysis with the same framework: two Xenium tissue-microarray slides (one lung, one liver) hosting 28 donors in total, processed with two organ-specific 439-gene panels (lung and liver). The lung TMA carries 16 donors — 10 with telomere-biology-disorder-associated pulmonary fibrosis plus 6 non-disease controls stratified by telomere status (non-short telomere) and neoplasia status (non-neoplastic) — and the liver TMA carries 12 donors — 4 with telomere-biology-disorder-associated hepatic fibrosis, 4 with alcoholic cirrhosis as a non-telomere fibrotic comparator, and 4 healthy controls. Each donor is represented by several technical-replicate cores (≥ 3); the unit of statistical replication is the donor and the unit of spatial inference is the core. Adapting the current pipeline therefore requires (i) externalization of hard-coded paths and tissue-specific marker sets (**P0**: done), (ii) parameterization of QC and clustering thresholds (**P0**:

done), (iii) TMA-aware multi-sample integration via scVI with sample-as-core and donor-
as-categorical-covariate (**P1**: done), (iv) dual-method pseudobulk differential expression
at the donor level (decoupler + muscat with consensus reporting; **P1**: done), and (v)
per-core Spacia followed by donor- and condition-level meta-analysis (**P1**: scaffolded). The
portability work is being prioritized over additional methodological refinements (e.g., SVG
null comparison, synthetic benchmarks against ground-truth ligand–receptor pairs) because
the cohort arrival date is fixed and the analytical contrast that distinguishes telomere-
disorder fibrosis from alcoholic cirrhosis requires the multi-sample contrast machinery to
be operational on day 1.

Supplementary Figure 1 — Quality control metrics

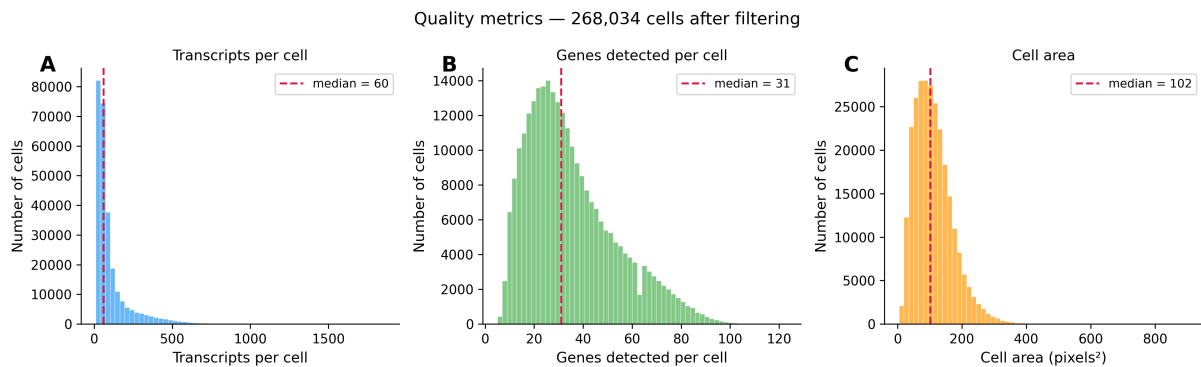


Figure S1: Quality control metrics. Distribution of per-cell transcript count, detected genes, and cell area before and after filtering. Thresholds applied: transcript count ≥ 5 , cell area $\leq 1,500 \mu\text{m}^2$. A total of 268,034 cells passed quality control from 278,659 detected.

Supplementary Figure 2 — Banksy versus Novae spatial-domain confusion matrix

Supplementary Figure 3 — Macrophage polarization-marker directionality along pseudotime

Supplementary Table S4 — Complete Spacia validation results for the 30 candidate LIANA+ interactions

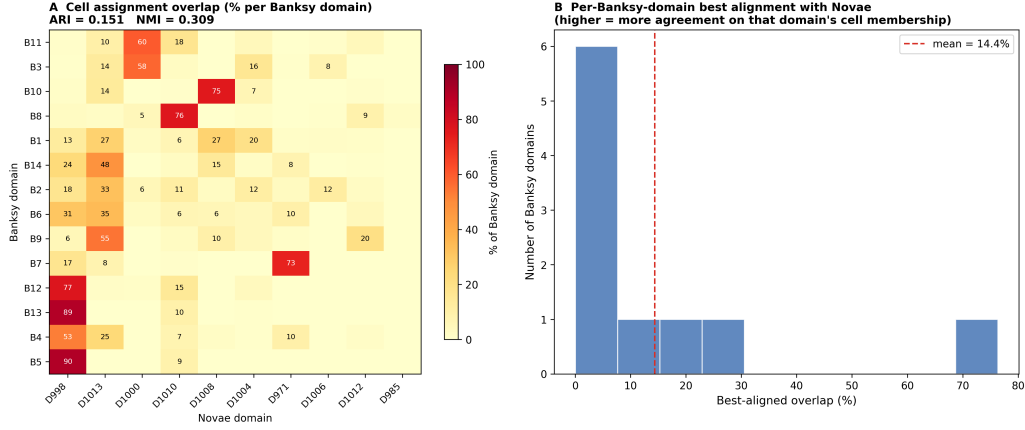


Figure S2: Confusion matrix between Banksy and Novae spatial-domain partitionings. Rows: Banksy domains (B1–B14, 14 fine-grained graph-based domains, $\lambda = 0.2$, $k_{\text{geom}} = 6$). Columns: Novae foundation-model domains (D971–D1013, 10 coarse compartments derived from 64-dimensional embeddings). Each cell shows the percentage of Banksy-domain cells that fall in the corresponding Novae domain. The matrix is asymmetric and far from block-diagonal, consistent with the partition-level disagreement quantified by ARI = 0.15 and NMI = 0.31. Several Banksy domains have an unambiguous Novae match (e.g., B5 $\rightarrow$ D998, 90%; B8 $\rightarrow$ D1010, 76%), supporting the strict per-cell consensus filter used to define the 35.4% high-confidence subset shown in Figure 3C; the remaining 64.6% concentrate at compartment boundaries where the two methods, by design, partition differently.

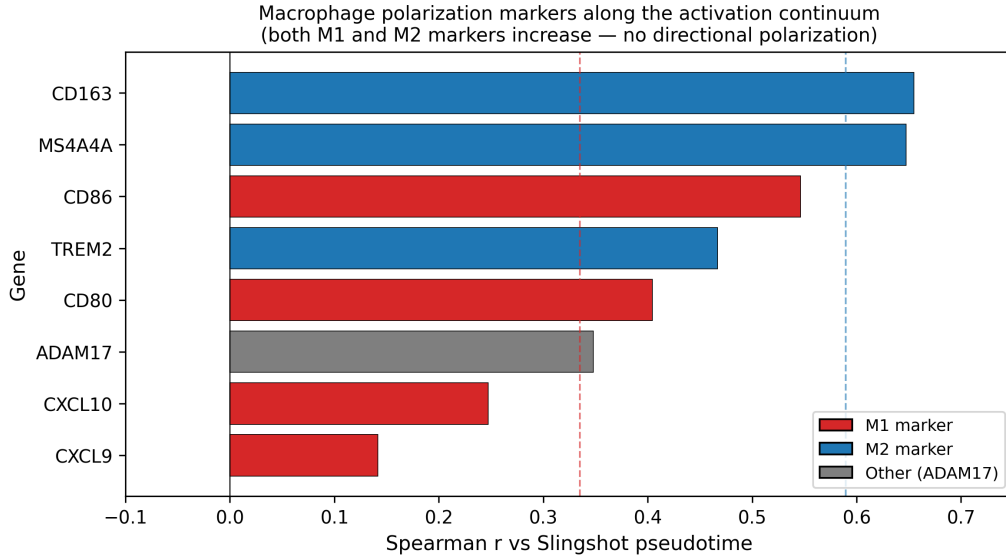


Figure S3: Polarization markers correlate with pseudotime in a non-directional pattern. Spearman rank correlation r between Slingshot pseudotime and the expression of canonical M1 (CD80, CD86, CXCL9, CXCL10) and M2 (CD163, MS4A4A, TREM2) markers, plus ADAM17, computed across the 22,902-cell macrophage compartment. If the pseudotime axis represented a directional M1 $\rightarrow$ M2 polarization, M1 markers would yield negative correlations and M2 markers positive correlations. Instead, *both* classes show positive correlations (M1: mean $r = +0.34$, range $+0.14$ to $+0.55$; M2: mean $r = +0.59$, range $+0.47$ to $+0.65$), with M2 markers rising more steeply. This is the empirical evidence cited in Results 3.5 against a directional polarization model and in favor of a global-activation continuum.

Sender	Receiver	Ligand	Receptor	β	Spacia p_{adj}	Validated
AT1	Tumor_resting	ADAM17	MUC1	4.060	1.10×10^{-71}	Yes
Tumor_resting	AT1	LTF	AGER	-4.297	9.37×10^{-66}	Yes
Monocyte_classical	Tumor_resting	ADAM17	MUC1	2.845	8.44×10^{-55}	Yes
Pericyte	Tumor_resting	ADAM17	MUC1	1.011	3.51×10^{-48}	Yes
Macrophage_M1	Epithelial_general	ADAM17	MUC1	2.288	5.91×10^{-43}	Yes
Epithelial_general	Endothelial_lymphatic	CDH1	EGFR	2.184	6.64×10^{-42}	Yes
Epithelial_general	AT1	CDH1	EGFR	1.939	1.18×10^{-32}	Yes
cDC2	Tumor_resting	ADAM17	MUC1	1.839	4.97×10^{-31}	Yes
Macrophage_M1	Tumor_resting	ADAM17	MUC1	1.525	4.92×10^{-28}	Yes
Macrophage_M2	Tumor_resting	ADAM17	MUC1	1.303	1.35×10^{-24}	Yes
cDC1	AT1	S100B	AGER	1.302	1.89×10^{-24}	Yes
Epithelial_general	Ciliated	CDH1	EGFR	-0.926	6.32×10^{-15}	Yes
cDC2	AT1	S100B	AGER	0.968	4.01×10^{-14}	Yes
Tumor_proliferating	AT1	S100B	AGER	-0.814	4.01×10^{-11}	Yes
pDC	Epithelial_general	ADAM17	MUC1	-0.854	1.26×10^{-10}	Yes
Epithelial_general	Plasmablast	CDH1	EGFR	-0.525	6.66×10^{-9}	Yes
Monocyte_classical	Epithelial_general	ADAM17	MUC1	0.639	1.82×10^{-8}	Yes
Epithelial_general	Plasma	CDH1	EGFR	-0.562	2.91×10^{-8}	Yes
Epithelial_general	Tumor_resting	CDH1	EGFR	-0.005	6.27×10^{-7}	Yes
Ciliated	AT1	LTF	AGER	-0.233	1.17×10^{-4}	Yes
cDC1	Tumor_resting	ADAM17	MUC1	0.172	0.048	Per-test
Ciliated	Endothelial_blood	CD24	SELP	0.220	0.049	Per-test
pDC	AT1	S100B	AGER	-0.274	0.068	No
Epithelial_general	Tumor_resting	ADAM17	MUC1	0.033	0.448	No
Pericyte	Epithelial_general	ADAM17	MUC1	-0.430	0.545	No
Plasma	Tumor_resting	ADAM17	MUC1	—	—	N/A
pDC	Tumor_resting	ADAM17	MUC1	—	—	N/A
Plasmablast	Tumor_resting	ADAM17	MUC1	—	—	N/A
Plasma	Epithelial_general	ADAM17	MUC1	—	—	N/A
DC_mature	Tumor_resting	ADAM17	MUC1	—	—	N/A

Table 1: Complete Spacia validation table for the 30 LIANA+ candidate interactions.

Columns: sender and receiver cell types; ligand and receptor (first listed gene for multi-subunit complexes, see Methods); Spacia pathway-level coefficient β (sign indicates spatial enrichment direction); Spacia adjusted p -value at the pathway level; dual-validation outcome under Bonferroni correction across the 30 cross-tests ($p_{\text{adj}} < 1.67 \times 10^{-3}$). Five rows have empty β / p -value cells: these correspond to Spacia fits that did not complete because the sender cell type contained fewer than 500 cells (Plasma, Plasmablast, mature DC, and pDC interactions); they are reported transparently as inconclusive rather than excluded a priori. Per-row source: `human_lung_cancer/results/02_biology/phase_f_spacia/F3_spacia_results.csv`.